

Spontaneous Parametric Down Conversion (SPDC): Maximizing Pair Rate for Type-II SPDC With PPKTP

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Entangled photons are essential for quantum communication and computation, where high heralding efficiency—i.e., the likelihood of detecting a photon given its entangled counterpart is detected—is critical for reliable operations [1,2]. Spontaneous Parametric Down Conversion (SPDC), a second-order nonlinear optical process (χ^2), is the most common technique for generating such photons. The efficiency and generation rate depend upon factors like crystal type, crystal length, pump power, and the beam waists of the pump, signal and idler.

In this work, we theoretically modelled the optimal parameters for high photon pair generation using a type-II SPDC process in a periodically poled KTiOPO_4 (PPKTP) crystal with a poling period of $10\mu\text{m}$ and a length of 2mm . Based on quasi-phase matching conditions, (i) $K_p = K_s \cos(\theta_s) + K_i \cos(\theta_i) + 2\pi/\Lambda$ and (ii) $\omega_p = \omega_s + \omega_i$, with $K_j = 2\pi n_j / \lambda_j$ and refractive indices n_j given by temperature-dependent Sellmeier equations [3–5]—we found the quasi-phase matching temperature for degenerate, collinear case to be 40.66°C for a 405 nm pump. Fig. 1 (a) shows how the quasi-phase matching temperature varies with SPDC emission angle.

Now, we proceed to calculate the photon pair generation rate and heralding efficiency in a non-collinear, degenerate SPDC setup, with emission angles $\theta_s = \theta_i = 3^\circ$. For such small angles ($\leq 3^\circ$), the effects of the Poynting vector and spatial walk-off can be neglected. The method outlined by Rockovich et. al. [6] is used to perform these calculations. Photon pair rates (in pairs/s.mW) are computed using the following expression:

$$R = \frac{\eta_s \eta_i P d_{eff}^2 \omega_s \omega_i}{\sqrt{2} \pi^{3/2} \epsilon_0 c^3 n_s n_i n_p B_p} \int_{-\infty}^{\infty} T_s(\omega_s) \int_{-\infty}^{\infty} T_i(\omega_p - \omega_s) |\Phi(\omega_s, \omega_i)|^2 d\omega_s d\omega_p \quad (1)$$

where η ($=1$ ideal) is efficiency of filters, B_p ($=30\text{THz}$) is pump beam spectral bandwidth, d_{eff} ($=2d_{24}/\pi$) is effective second order nonlinear coefficient, T_j are filter transmission functions (unit bandpass filter here) and Joint Spectral Amplitude Φ is given in [6]. We obtained highest pair rate of 2595.60 pairs/s.mW at pump beam waist of $119.22\mu\text{m}$ as shown in Fig. 1 (b). Further we show variation of pair rate with signal and idler beam waist [Fig. 1 (c)], power of pump beam [Fig. 1 (d)] and length of the crystal [Fig. 1 (e)].

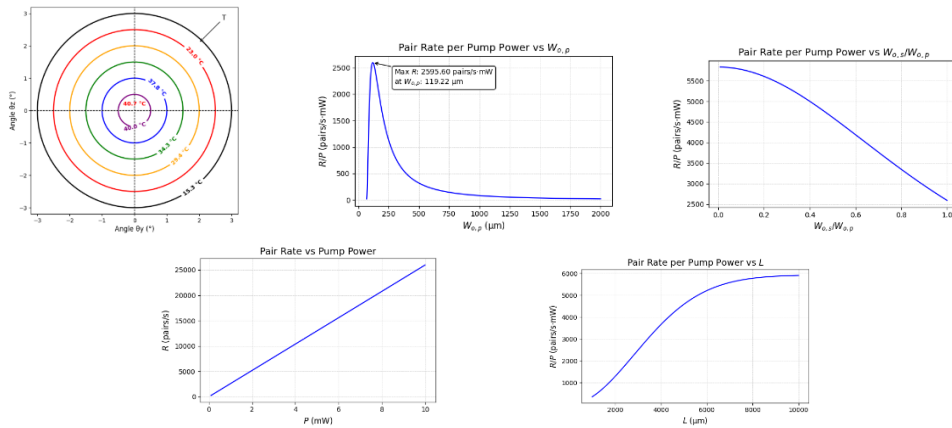


Fig. 1. (a) Quasi-phase matching temperature at different angles. **(b)** The pair production rate as a function of the pump beam waist. **(c)** Pair production rate as a function of $W_{0,s}/W_{0,p}$. **(d)** The pair production rate as a function of the power of the pump beam. **(e)** The pair production rate as a function of the length of the crystal.

References

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